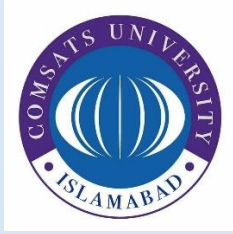


**COMSATS UNIVERSITY ISLAMABAD,
LAHORE CAMPUS**



SUBMITTED BY: GROUP A

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SUBMITTED TO: MISS RABAIL ASGHAR

PF-LAB SEMESTER PROJECT

**LIBRARY MANAGEMENT
SYSTEM**

BS ARTIFICIAL INTELLIGENCE (BAI-A-SP26)

PROGRAM STATEMENT

Dr Tariq Najmi is a librarian at a school and currently manages all library records manually.

To improve efficiency, he wants to computerise the system and has hired you to develop a

Library Management System using C++.

CONCEPTS

- Structures
- Pointers
- Functions
- File Handling

BOOK INFORMATION

1. Book Serial Number
2. ISBN Number
3. Book Title
4. Edition
5. Author Name
6. Is_Reserved (Issued/Reserved status)

REQUIREMENTS

Your system should provide the following menu options:

1. Add New Book
2. Delete Book
3. Search Book by:
 - ISBN NUMBER
 - BOOK TITLE
 - AUTHOR NAME
4. Modify/Update Book Record
5. Display All Books
6. Issue/Reserve a Book
7. Exit Program

CODE

```
/*
```

```
    BS ARTIFICIAL INTELLIGENCE (BAI-A-SP26)
```

```
    PF-LAB Semester 01 Project
```

```
    Library Management System
```

```
    Group A
```

```
    Members: Faryal Sarfraz (SP26-BAI-019), Hania Tanveer (SP26-BAI-024), Sharmeen Imtiaz  
    (SP26-BAI-045), and Taha Rizwan (SP26-BAI-049)
```

```
*/
```

```
#include <iostream>
```

```
#include <fstream>
```

```
#include <cstring>
```

```
#include <iomanip>
```

```
using namespace std;
```

```
// STRUCTURE DEFINITION
```

```
struct Book {
```

```
    int  serialNumber;
```

```
    char isbn[20];
```

```
    char title[100];
```

```
    int  edition;
```

```
    char author[60];
```

```
    bool isReserved; // 0 = available, 1 = issued/reserved
```

```
};
```

```
// CONSTANTS & GLOBALS
```

```
const char FILE_NAME[] = "library.txt";
```

```
// FUNCTION PROTOTYPES
```

```
void displayMenu();
```

```
void addBook();
```

```
void deleteBook();
```

```
void searchMenu();
```

```
void searchByISBN();
```

```
void searchByTitle();
```

```
void searchByAuthor();
```

```
void modifyBook();
```

```
void displayAllBooks();
```

```
void issueReserveBook();
```

```
void printBookHeader();
```

```
void printBookRow(const Book* b);
```

```
void pause();
```

```
// MAIN
```

```
int main() {
```

```
    int choice;
```

```
    cout << "\n =====";
```

```
    cout << "\n ||  LIBRARY MANAGEMENT SYSTEM  ||";
```

```
    cout << "\n ||  Dr. Tariq Najmi's Library  ||";
```

```
    cout << "\n =====\n";
```

```
    do {
```

```
        displayMenu();
```

```
        cout << " Enter your choice: ";
```

```

cin >> choice;

cin.ignore();

switch (choice) {
    case 1: addBook();    break;
    case 2: deleteBook(); break;
    case 3: searchMenu(); break;
    case 4: modifyBook(); break;
    case 5: displayAllBooks(); break;
    case 6: issueReserveBook(); break;
    case 7:
        cout << "\n Thank you for using the Library System. Goodbye!\n\n";
        break;
    default:
        cout << "\n Invalid choice. Please try again.\n";
}
} while (choice != 7);

return 0;
}

// MENU
void displayMenu() {
    cout << "\n =====\n";
    cout << " ||      MAIN MENU      ||\n";
    cout << " =====\n";
    cout << " | 1. Add New Book      |\n";

```

```

cout << " | 2. Delete Book      |\n";
cout << " | 3. Search Book      |\n";
cout << " | 4. Modify/Update Book  |\n";
cout << " | 5. Display All Books   |\n";
cout << " | 6. Issue / Reserve a Book |\n";
cout << " | 7. Exit                |\n";
cout << " =====\n";
}

```

// Print table header

```

void printBookHeader() {
    cout << "\n " << string(95, '-') << "\n";
    cout << " " << left
        << setw(6) << "S.No"
        << setw(16) << "ISBN"
        << setw(30) << "Title"
        << setw(5) << "Ed."
        << setw(25) << "Author"
        << setw(12) << "Status"
        << "\n";
    cout << " " << string(95, '-') << "\n";
}

```

// Print one book row

```

void printBookRow(const Book* b) {
    cout << " " << left
        << setw(6) << b->serialNumber

```

```
        << setw(16) << b->isbn
        << setw(30) << b->title
        << setw(5) << b->edition
        << setw(25) << b->author
        << setw(12) << (b->isReserved ? "Reserved" : "Available")
        << "\n";
    }
}
```

```
// Press Enter to continue
```

```
void pause() {
    cout << "\n Press Enter to continue...";
    cin.get();
}
```

```
// 1. ADD NEW BOOK
```

```
void addBook() {
    Book newBook;
    newBook.isReserved = false;

    cout << "\n || ADD NEW BOOK ||\n";
    cout << " Enter Serial Number: ";
    cin >> newBook.serialNumber;
    cin.ignore();

    cout << " ISBN : ";
    cin.getline(newBook.isbn, 20);
}
```

```
cout << " Title : ";
cin.getline(newBook.title, 100);

cout << " Edition : ";
cin >> newBook.edition;
cin.ignore();

cout << " Author Name : ";
cin.getline(newBook.author, 60);

// Append to text file
ofstream fout(FILE_NAME, ios::app);
if (!fout) {
    cout << "\n Error opening file.\n";
    return;
}

// Write data fields separated by newlines
fout << newBook.serialNumber << "\n"
    << newBook.isbn << "\n"
    << newBook.title << "\n"
    << newBook.edition << "\n"
    << newBook.author << "\n"
    << newBook.isReserved << "\n";

fout.close();
```

```
    cout << "\n Book added successfully! Serial No: " << newBook.serialNumber << "\n";  
    pause();  
}
```

```
// 2. DELETE BOOK
```

```
void deleteBook() {  
    ifstream fin(FILE_NAME);  
    if (!fin) {  
        cout << "\n No records found.\n";  
        pause();  
        return;  
    }  
}
```

```
int targetSerial;  
cout << "\n || DELETE BOOK ||\n";  
cout << " Enter Serial Number to delete: ";  
cin >> targetSerial;  
cin.ignore();
```

```
ofstream fout("temp.txt");  
Book b;  
bool found = false;
```

```
while (fin >> b.serialNumber) {  
    fin.ignore();  
    fin.getline(b.isbn, 20);  
    fin.getline(b.title, 100);
```

```

fin >> b.edition;

fin.ignore();

fin.getline(b.author, 60);

fin >> b.isReserved;

fin.ignore();


if (b.serialNumber == targetSerial) {

    found = true;

    cout << "\n Book found:\n";

    printBookHeader();

    printBookRow(&b);

    cout << " " << string(95, '-') << "\n";

    cout << " Confirm delete? (y/n): ";

    char confirm;

    cin >> confirm;

    cin.ignore();

    if (confirm == 'y' || confirm == 'Y') {

        cout << " Book deleted.\n";

        continue; // skip saving it to temp file

    } else {

        cout << " Deletion cancelled.\n";

    }

}


// Write kept records back to temporary file

fout << b.serialNumber << "\n" << b.isbn << "\n" << b.title << "\n"

    << b.edition << "\n" << b.author << "\n" << b.isReserved << "\n";

```

```

    }

    fin.close();
    fout.close();

    remove(FILE_NAME);
    rename("temp.txt", FILE_NAME);

    if (!found)
        cout << "\n Book with Serial No " << targetSerial << " not found.\n";

    pause();
}

// 3. SEARCH MENU
void searchMenu() {
    int choice;

    cout << "\n || SEARCH BOOK BY ||\n";
    cout << " 1. ISBN Number\n";
    cout << " 2. Book Title\n";
    cout << " 3. Author Name\n";
    cout << " Enter choice: ";

    cin >> choice;
    cin.ignore();

    switch (choice) {
        case 1: searchByISBN(); break;
    }
}

```

```
        case 2: searchByTitle(); break;
        case 3: searchByAuthor(); break;
        default: cout << " Invalid choice.\n";
    }
}
```

```
void searchByISBN() {
    ifstream fin(FILE_NAME);
    if (!fin) { cout << "\n No records found.\n"; pause(); return; }

    char key[20];
    cout << " Enter ISBN to search: ";
    cin.getline(key, 20);

    Book b;
    bool found = false;

    printBookHeader();
    while (fin >> b.serialNumber) {
        fin.ignore();
        fin.getline(b.isbn, 20);
        fin.getline(b.title, 100);
        fin >> b.edition;
        fin.ignore();
        fin.getline(b.author, 60);
        fin >> b.isReserved;
        fin.ignore();
```

```

        if (strcmp(b.isbn, key) == 0) {
            printBookRow(&b);
            found = true;
        }
    }
    fin.close();
    if (!found) cout << " No book found with ISBN: " << key << "\n";
    cout << " " << string(95, '-') << "\n";
    pause();
}

```

```

void searchByTitle() {
    ifstream fin(FILE_NAME);
    if (!fin) { cout << "\n No records found.\n"; pause(); return; }

    char key[100];
    cout << " Enter Title to search: ";
    cin.getline(key, 100);

    Book b;
    bool found = false;

    printBookHeader();
    while (fin >> b.serialNumber) {
        fin.ignore();
        fin.getline(b.isbn, 20);
        fin.getline(b.title, 100);
    }
}

```

```

    fin >> b.edition;

    fin.ignore();

    fin.getline(b.author, 60);

    fin >> b.isReserved;

    fin.ignore();


    if (strcmp(b.title, key) == 0) {
        printBookRow(&b);
        found = true;
    }
}

fin.close();

if (!found) cout << " No book found with title: " << key << "\n";

cout << " " << string(95, '-') << "\n";

pause();

}

void searchByAuthor() {
    ifstream fin(FILE_NAME);

    if (!fin) { cout << "\n No records found.\n"; pause(); return; }

    char key[60];

    cout << " Enter Author Name to search: ";

    cin.getline(key, 60);

    Book b;

    bool found = false;

```

```

printBookHeader();
while (fin >> b.serialNumber) {
    fin.ignore();
    fin.getline(b.isbn, 20);
    fin.getline(b.title, 100);
    fin >> b.edition;
    fin.ignore();
    fin.getline(b.author, 60);
    fin >> b.isReserved;
    fin.ignore();

    if (strcmp(b.author, key) == 0) {
        printBookRow(&b);
        found = true;
    }
}
fin.close();
if (!found) cout << " No book found by author: " << key << "\n";
cout << " " << string(95, '-') << "\n";
pause();
}

```

// 4. MODIFY / UPDATE BOOK

```

void modifyBook() {
    ifstream fin(FILE_NAME);
    if (!fin) { cout << "\n No records found.\n"; pause(); return; }
}

```

```
int targetSerial;

cout << "\n || MODIFY BOOK ||\n";

cout << " Enter Serial Number to modify: ";

cin >> targetSerial;

cin.ignore();
```

```
ofstream fout("temp.txt");

Book b;

bool found = false;
```

```
while (fin >> b.serialNumber) {

    fin.ignore();

    fin.getline(b.isbn, 20);

    fin.getline(b.title, 100);

    fin >> b.edition;

    fin.ignore();

    fin.getline(b.author, 60);

    fin >> b.isReserved;

    fin.ignore();
```

```
if (b.serialNumber == targetSerial) {

    found = true;

    cout << "\n Current Record:\n";

    printBookHeader();

    printBookRow(&b);

    cout << " " << string(95, '-') << "\n";
```

```

cout << "\n Enter new details (press Enter to keep current):\n";
char temp[100];

cout << " ISBN [" << b.isbn << "]: ";
cin.getline(temp, 20);
if (strlen(temp) > 0) strcpy(b.isbn, temp);

cout << " Title [" << b.title << "]: ";
cin.getline(temp, 100);
if (strlen(temp) > 0) strcpy(b.title, temp);

cout << " Edition [" << b.edition << "]: ";
cin.getline(temp, 10);
if (strlen(temp) > 0) {
    b.edition = 0;
    for(int i=0; temp[i] >= '0' && temp[i] <= '9'; i++) {
        b.edition = b.edition * 10 + (temp[i] - '0');
    }
}

cout << " Author [" << b.author << "]: ";
cin.getline(temp, 60);
if (strlen(temp) > 0) strcpy(b.author, temp);

cout << "\n Book updated successfully.\n";
}

```

```

        fout << b.serialNumber << "\n" << b.isbn << "\n" << b.title << "\n"
            << b.edition << "\n" << b.author << "\n" << b.isReserved << "\n";
    }

    fin.close();
    fout.close();

    remove(FILE_NAME);
    rename("temp.txt", FILE_NAME);

    if (!found) cout << "\n Book with Serial No " << targetSerial << " not found.\n";
    pause();
}

// 5. DISPLAY ALL BOOKS
void displayAllBooks() {
    ifstream fin(FILE_NAME);
    if (!fin) { cout << "\n No records found.\n"; pause(); return; }

    Book b;
    int count = 0;

    cout << "\n || ALL BOOKS ||";
    printBookHeader();

    while (fin >> b.serialNumber) {
        fin.ignore();

```

```

        fin.getline(b.isbn, 20);
        fin.getline(b.title, 100);
        fin >> b.edition;
        fin.ignore();
        fin.getline(b.author, 60);
        fin >> b.isReserved;
        fin.ignore();

        printBookRow(&b);
        count++;
    }
    fin.close();

    cout << " " << string(95, '-') << "\n";
    cout << " Total Books: " << count << "\n";
    pause();
}

// 6. ISSUE / RESERVE A BOOK
void issueReserveBook() {
    ifstream fin(FILE_NAME);
    if (!fin) { cout << "\n No records found.\n"; pause(); return; }

    int targetSerial;
    cout << "\n || ISSUE / RESERVE BOOK ||\n";
    cout << " Enter Serial Number: ";
    cin >> targetSerial;
    cin.ignore();

```

```
ofstream fout("temp.txt");
```

```
Book b;
```

```
bool found = false;
```

```
while (fin >> b.serialNumber) {
```

```
    fin.ignore();
```

```
    fin.getline(b.isbn, 20);
```

```
    fin.getline(b.title, 100);
```

```
    fin >> b.edition;
```

```
    fin.ignore();
```

```
    fin.getline(b.author, 60);
```

```
    fin >> b.isReserved;
```

```
    fin.ignore();
```

```
if (b.serialNumber == targetSerial) {
```

```
    found = true;
```

```
    cout << "\n Book: " << b.title << " by " << b.author << "\n";
```

```
    cout << " Current Status: " << (b.isReserved ? "Reserved/Issued" : "Available") << "\n\n";
```

```
if (b.isReserved) {
```

```
    cout << " This book is currently reserved. Return it? (y/n): ";
```

```
    char c; cin >> c; cin.ignore();
```

```
    if (c == 'y' || c == 'Y') {
```

```
        b.isReserved = false;
```

```
        cout << " Book returned. Status: Available\n";
```

```
    } else {
```

```

        cout << " No changes made.\n";
    }
} else {
    cout << " Issue or Reserve this book? (y/n): ";
    char c; cin >> c; cin.ignore();
    if (c == 'y' || c == 'Y') {
        b.isReserved = true;
        cout << " Book issued/reserved successfully.\n";
    } else {
        cout << " No changes made.\n";
    }
}
}

fout << b.serialNumber << "\n" << b.isbn << "\n" << b.title << "\n"
    << b.edition << "\n" << b.author << "\n" << b.isReserved << "\n";
}

fin.close();
fout.close();

remove(FILE_NAME);
rename("temp.txt", FILE_NAME);

if (!found) cout << "\n Book with Serial No " << targetSerial << " not found.\n";
pause();
}

```

OUTPUTS (TEST CASES)

1. ADD NEW BOOK

```
C:\Users\DO Star Asia\Downloads\library.exe

=====
||  LIBRARY MANAGEMENT SYSTEM  ||
||  Dr. Tariq Najmi's Library  ||
=====

=====
||          MAIN MENU          ||
=====
| 1. Add New Book              |
| 2. Delete Book               |
| 3. Search Book               |
| 4. Modify/Update Book        |
| 5. Display All Books         |
| 6. Issue / Reserve a Book    |
| 7. Exit                      |
=====
Enter your choice: 1

|| ADD NEW BOOK ||
Enter Serial Number: 1
ISBN : 7654
Title : DUNE
Edition : 6
Author Name : FRANK
Book added successfully! Serial No: 1
Press Enter to continue...

C:\Users\DO Star Asia\Downloads\library.exe

=====
||          MAIN MENU          ||
=====
| 1. Add New Book              |
| 2. Delete Book               |
| 3. Search Book               |
| 4. Modify/Update Book        |
| 5. Display All Books         |
| 6. Issue / Reserve a Book    |
| 7. Exit                      |
=====
Enter your choice: 1

|| ADD NEW BOOK ||
Enter Serial Number: 2
ISBN : 4444
Title : CRIME AND PUNISHMENT
Edition : 3
Author Name : FYODOR
Book added successfully! Serial No: 2
Press Enter to continue...

C:\Users\DO Star Asia\Downloads\library.exe

=====
||          MAIN MENU          ||
=====
| 1. Add New Book              |
| 2. Delete Book               |
| 3. Search Book               |
| 4. Modify/Update Book        |
| 5. Display All Books         |
| 6. Issue / Reserve a Book    |
| 7. Exit                      |
=====
Enter your choice: 1

|| ADD NEW BOOK ||
Enter Serial Number: 3
ISBN : 55555
Title : FUNDAMENTALS OF PHYSICS VOL1
Edition : 3
Author Name : DAVID
Book added successfully! Serial No: 3
Press Enter to continue...
```

2. DELETE BOOK

DELETED

```
Press Enter to continue...

=====
||      MAIN MENU      ||
=====
| 1. Add New Book      |
| 2. Delete Book       |
| 3. Search Book       |
| 4. Modify/Update Book|
| 5. Display All Books  |
| 6. Issue / Reserve a Book|
| 7. Exit              |
=====

Enter your choice: 2

|| DELETE BOOK ||
Enter Serial Number to delete: 2

Book found:

-----
S.No  ISBN          Title                      Ed.  Author      Status
-----
2     4444          CRIME AND PUNISHMENT          3    FYODOR      Available
-----

Confirm delete? (y/n): Y
Book deleted.

Press Enter to continue...
```

NOT FOUND

```
=====
||      MAIN MENU      ||
=====
| 1. Add New Book      |
| 2. Delete Book       |
| 3. Search Book       |
| 4. Modify/Update Book|
| 5. Display All Books  |
| 6. Issue / Reserve a Book|
| 7. Exit              |
=====

Enter your choice: 2

|| DELETE BOOK ||
Enter Serial Number to delete: 8

Book with Serial No 8 not found.

Press Enter to continue...
```

3. SEARCH BOOK

ISBN NUMBER (FOUND)

```
=====
||      MAIN MENU      ||
=====
| 1. Add New Book      |
| 2. Delete Book       |
| 3. Search Book       |
| 4. Modify/Update Book|
| 5. Display All Books |
| 6. Issue / Reserve a Book|
| 7. Exit              |
=====
Enter your choice: 3

|| SEARCH BOOK BY ||
1. ISBN Number
2. Book Title
3. Author Name
Enter choice: 1
Enter ISBN to search: 7654

-----
S.No  ISBN          Title                Ed.  Author          Status
-----
1      7654          DUNE                6    FRANK            Available
-----

Press Enter to continue...
```

BOOK TITLE (FOUND)

```
=====
||      MAIN MENU      ||
=====
| 1. Add New Book      |
| 2. Delete Book       |
| 3. Search Book       |
| 4. Modify/Update Book|
| 5. Display All Books |
| 6. Issue / Reserve a Book|
| 7. Exit              |
=====
Enter your choice: 3

|| SEARCH BOOK BY ||
1. ISBN Number
2. Book Title
3. Author Name
Enter choice: 2
Enter Title to search: ANIMAL FARM

-----
S.No  ISBN          Title                Ed.  Author          Status
-----
5      877560          ANIMAL FARM         8    GEORGE           Available
-----

Press Enter to continue...
```

4. MODIFY/UPDATE BOOK

C:\Users\DO Star Asia\Downloads\library.exe

- 3. Search Book
- 4. Modify/Update Book
- 5. Display All Books
- 6. Issue / Reserve a Book
- 7. Exit

Enter your choice: 4

|| MODIFY BOOK ||

Enter Serial Number to modify: 4

Current Record:

S.No	ISBN	Title	Ed.	Author	Status
4	777889	FUNDAMENTALS OF PHYSICS VOL2	3	DAVID	Available

Enter new details (press Enter to keep current):

ISBN [777889]:

Title [FUNDAMENTALS OF PHYSICS VOL2]:

Edition [3]: 4

Author [DAVID]:

Book updated successfully.

Press Enter to continue...

5. DISPLAY ALL BOOKS

|| MAIN MENU ||

- 1. Add New Book
- 2. Delete Book
- 3. Search Book
- 4. Modify/Update Book
- 5. Display All Books
- 6. Issue / Reserve a Book
- 7. Exit

Enter your choice: 5

|| ALL BOOKS ||

S.No	ISBN	Title	Ed.	Author	Status
1	7654	DUNE	6	FRANK	Available
3	55555	FUNDAMENTALS OF PHYSICS VOL1	3	DAVID	Available
4	777889	FUNDAMENTALS OF PHYSICS VOL2	4	DAVID	Available
5	877560	ANIMAL FARM	8	GEORGE	Available
6	0111	HARRY POTTER	8	ROWLING	Available

Total Books: 5

Press Enter to continue...

6. RESERVE A BOOK

RESERVED

```
=====
||          MAIN MENU          ||
=====
|  1. Add New Book              |
|  2. Delete Book              |
|  3. Search Book              |
|  4. Modify/Update Book       |
|  5. Display All Books        |
|  6. Issue / Reserve a Book   |
|  7. Exit                     |
=====
Enter your choice:
6

|| ISSUE / RESERVE BOOK ||
Enter Serial Number: 6

Book: HARRY POTTER by ROWLING
Current Status: Available

Issue or Reserve this book? (y/n): Y
Book issued/reserved successfully.

Press Enter to continue...
```

NOT FOUND

```
=====
||          MAIN MENU          ||
=====
|  1. Add New Book              |
|  2. Delete Book              |
|  3. Search Book              |
|  4. Modify/Update Book       |
|  5. Display All Books        |
|  6. Issue / Reserve a Book   |
|  7. Exit                     |
=====
Enter your choice: 6

|| ISSUE / RESERVE BOOK ||
Enter Serial Number: 9

Book with Serial No 9 not found.

Press Enter to continue...
```

ALREADY RESERVED AND UNRESERVE

```
=====
||          MAIN MENU          ||
=====
|  1. Add New Book             |
|  2. Delete Book              |
|  3. Search Book              |
|  4. Modify/Update Book       |
|  5. Display All Books        |
|  6. Issue / Reserve a Book   |
|  7. Exit                     |
=====
Enter your choice: 6

|| ISSUE / RESERVE BOOK ||
Enter Serial Number: 1

Book: DUNE by FRANK
Current Status: Reserved/Issued

This book is currently reserved. Return it? (y/n): Y
Book returned. Status: Available

Press Enter to continue..._
```

7. EXIT

```
=====
||          MAIN MENU          ||
=====
|  1. Add New Book             |
|  2. Delete Book              |
|  3. Search Book              |
|  4. Modify/Update Book       |
|  5. Display All Books        |
|  6. Issue / Reserve a Book   |
|  7. Exit                     |
=====
Enter your choice: 7

Thank you for using the Library System. Goodbye!

-----
Process exited after 746.8 seconds with return value 0
Press any key to continue . . .
```